

No Tutorial Today

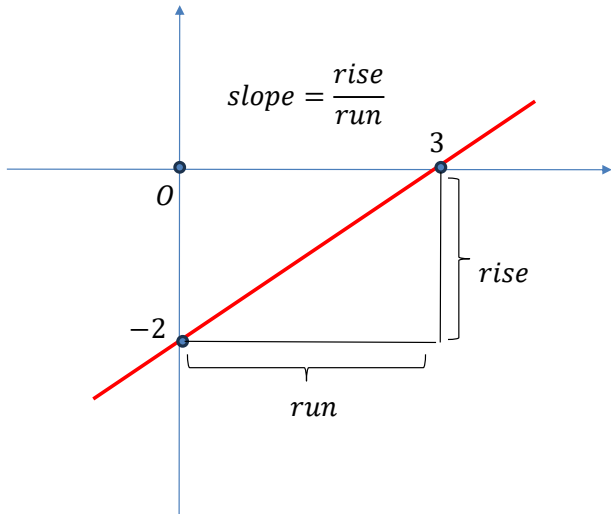
Online Quiz 1 will be available on Brightspace today at 5:30pm, and is due on Sep 20, 11pm.

Tutorials start on Sep 24 (next Thursday). Please go to your tutorial room according to the **last four digits** of your student number shown below:

Last four digits of student number	Tutorial section and room	TA name
0034-1827	Section B1, Room LA A720	Aaron Dwyer
1832-5267	Section B2, Room SA 406	Polina Vardashko
5318-7827	Section B3, Room NN 279	Biying Guo
7887-9914	Section B4, Room NN 280	Christopher Middlebrook

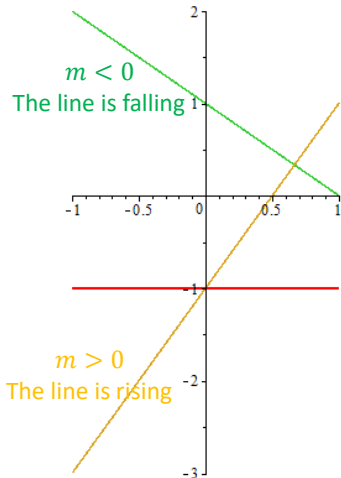
The Graph of Equation $2x - 3y = 6$

Its slope-intercept form is $y = \frac{2}{3}x - 2$, where $\frac{2}{3}$ is the slope and -2 is the y -intercept.



The Slope-Intercept Equation of a Line

In general, a linear equation in slope-intercept form is written as $y = mx + b$, where the constant m is the slope, and the constant b is the y -intercept.



$$y = mx + b$$

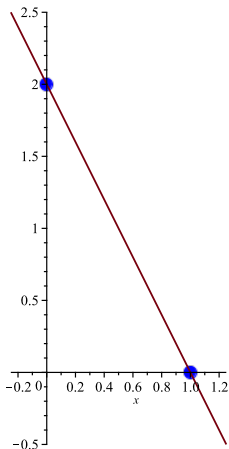
m is the slope,
 b is the y -intercept,
or vertical intercept.

$m = 0$
The line is horizontal

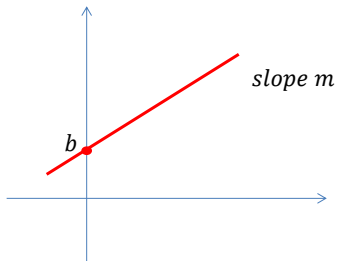
Examples

Example Consider the linear equation $2x + y = 2$.

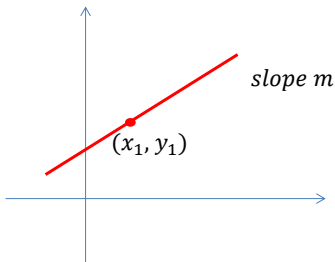
Its slope-intercept form is $y = -2x + 2$. Hence its graph is a line with slope -2 and y -intercept 2 .



The Point-Slope Equation of a Line

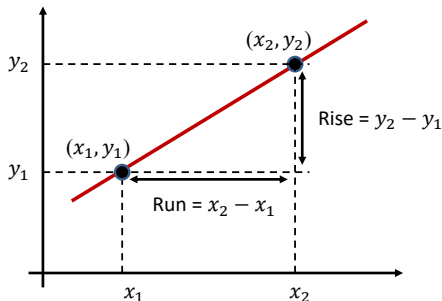


The *slope-intercept* equation:
$$y = mx + b$$



The *point-slope* equation of
the line is:
$$y = m(x - x_1) + y_1$$

The Two-Point Equation of a Line



The slope m of the line through two points (x_1, y_1) and (x_2, y_2) is given by the formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$.

The *two-point* equation of the line is:

$$y = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) + y_1$$

Examples

Example Find an equation of the line passing through points $(-3, 2)$ and $(1, 5)$.

Solution The slope of the line is equal to

$$\frac{5 - 2}{1 - (-3)} = \frac{3}{4}.$$

The point-slope equation of the line is

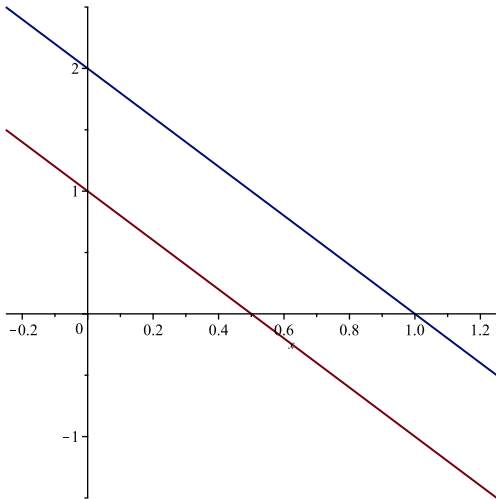
$$y = \frac{3}{4}(x - 1) + 5.$$

The slope-intercept equation of the line is

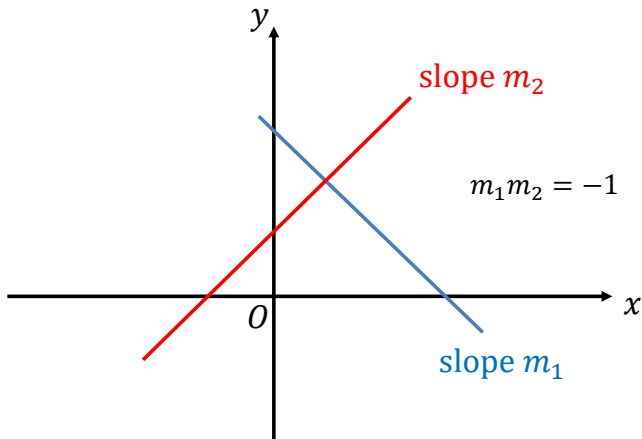
$$y = \frac{3}{4}x + \frac{17}{4}.$$

Parallel Lines

Example Parallel lines have the same slope.



Perpendicular Lines



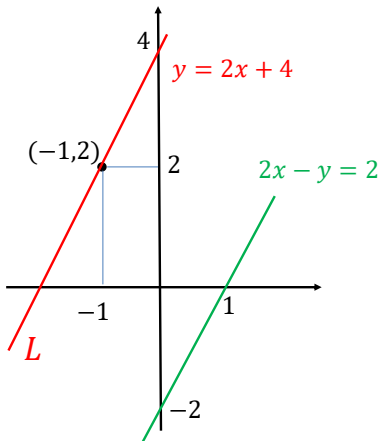
*Two lines are perpendicular to each other
if the product of their slopes is equal to -1*

Examples

Example Find an equation of the line L which passes through point $(-1, 2)$ and is parallel to the line $2x - y = 2$.

Solution The slope of line L is 2, which gives the following equation of L :

$$y = 2(x - (-1)) + 2, \text{ or } y = 2x + 4.$$

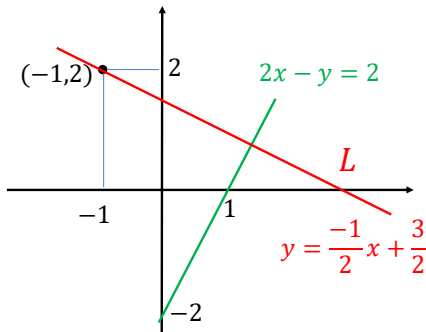


Examples

Example Find an equation of the line L which passes through point $(-1, 2)$ and is perpendicular to the line $2x - y = 2$.

Solution The slope of line L is $-1/2$, which gives the following equation of L :

$$y = \frac{-1}{2}(x - (-1)) + 2, \text{ or } y = \frac{-1}{2}x + \frac{3}{2}.$$

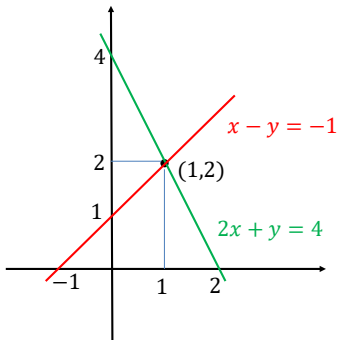


1.3 Systems of Linear Equations in Two Variables

Example Solve the following system of equations:

$$\begin{cases} x - y = -1 \\ 2x + y = 4 \end{cases}$$

Graphing Method: We want to find (x, y) which satisfies both equations. Graphically, we want to find the point of intersection of the two lines representing the two linear equations, as shown below.



1.3 Systems of Linear Equations in Two Variables

The point of intersection is $(1, 2)$, and we can verify that $x = 1, y = 2$ indeed is a solution:

$$\begin{cases} 1 - 2 = -1 \\ 2 \times 1 + 2 = 4 \end{cases}$$

Substitution Method: From the first equation in the original system we obtain

$$y = x + 1. \tag{1}$$

Substituting this into the second equation, we obtain

$$2x + (x + 1) = 4 \implies 3x = 3 \implies x = 1.$$

Substituting $x = 1$ into (1), we obtain $y = 2$.

1.3 Systems of Linear Equations in Two Variables

Elimination Method Adding the two equations (eliminate the variable y), we obtain

$$3x = 3 \implies x = 1.$$

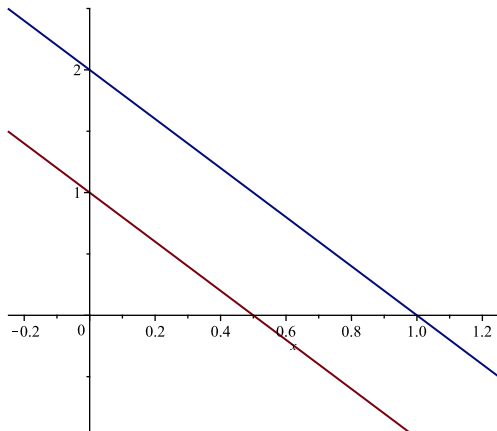
We may also eliminate the variable x as follows. Multiplying the first equation by 2 and then subtracting the second equation:

$$-2y - y = -2 - 4 \implies -3y = -6 \implies y = 2.$$

Using the first equation, we obtain $x = 2 - 1 = 1$.

1.3 Systems of Linear Equations in Two Variables

Example A system of two linear equations has no solution when the corresponding lines are parallel. That is, they are not identical lines and have the same slope.



1.3 Systems of Linear Equations in Two Variables

Example A system of two linear equations has infinitely many solutions when the corresponding lines are identical.

$$\begin{cases} x - y = -1 \\ -2x + 2y = 2 \end{cases}$$

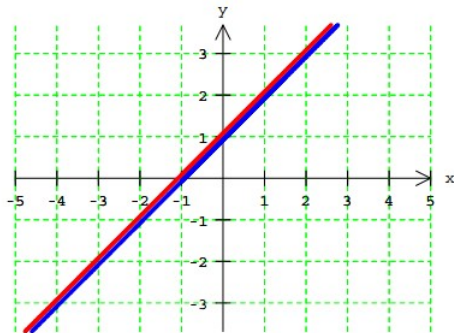


Figure 1.7: Lines $x - y = -1$ and $2x - 2y = -2$ coincide.

Systems of Linear Equations in Three Variables

Example Solve the following system of equations:

$$\begin{cases} 2x - 3y - 2z = 1, \\ 3x + y - z = 2, \\ 4x - y - 3z = 0. \end{cases}$$

Solution We may use the elimination method as follows.

Adding the last two equations, we obtain

$$7x - 4z = 2. \tag{2}$$

Multiplying the second equation by 3 and then add it to the first equation, we obtain

$$11x - 5z = 7. \tag{3}$$

Systems of Linear Equations in Three Variables

Now equations (2) and (3) contain only two variables x and z , which can be solved as before.

$$5 \times Eq(2) - 4 \times Eq(3) \implies -9x = -18 \implies x = 2.$$

Substituting $x = 2$ into (1), we obtain $z = 3$.

Substituting $x = 2, z = 3$ into the top equation of the original system, we obtain

$$-3y = 1 - 4 + 6 \implies y = -1.$$

Check that $x = 2, y = -1, z = 3$ indeed satisfy the original system.